Lecture 8-9

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Contents

1 Real Numbers and Decimal Representations **2**
1.1 Uniqueness of Decimal Representation 2
1.1.1 Theorem (Non-uniqueness Condition) 2
1.1.2 Proof of General Case 2
1.2 Periodic Decimals and Rational Numbers 3
1.2.1 Theorem 1: Rational $\implies$ Periodic 3
1.2.2 Theorem 2: Periodic $\implies$ Rational 4

2 Numeral Systems **4**
2.1 Positional Systems 4
2.1.1 Examples 4
2.1.2 Theorem (Existence and Uniqueness) 5
2.2 Non-Positional Systems 5

3 Number Theory Basics **5**
3.1 Divisibility and The Division Algorithm 5
3.2 Prime Numbers 5
3.3 Greatest Common Divisor (GCD) 5

4 Euclid's Algorithm **6**
4.1 The Procedure 6
4.1.1 Example 6
4.2 Correctness Lemma 6

5 Bézout's Identity (Extended Euclidean Algorithm) **7**
5.1 Theorem 7
5.1.1 Example 7
5.2 Proof 1: Constructive (Back-substitution) 7

5.3 Proof 2: Non-constructive (Well-Ordering Principle) 7

1 Real Numbers and Decimal Representations

1.1 Uniqueness of Decimal Representation

Problem: Is the decimal representation of a real number unique?

Consider $x \in (0, 1)$.

$$x = 0.a_1a_2a_3\ldots$$

$$x = 0.b_1b_2b_3\ldots$$

If these are two different representations of the same number, how do they differ?

1.1.1 Theorem (Non-uniqueness Condition)

If a real number has two different decimal representations, then one of them ends in an infinite sequence of 0s and the other in an infinite sequence of 9s.

Example:

$$0.5000\cdots = 0.4999\ldots$$

$$0.2600\cdots = 0.25999\ldots$$

Proof of $0.999\cdots = 1$: Using the geometric series formula $\sum_{n=0}^{\infty} ar^n = \frac{a}{1-r}$:

$$0.999\cdots = 9\cdot 10^{-1} + 9\cdot 10^{-2} + \cdots = \sum_{n=1}^{\infty} \frac{9}{10^n} = \frac{9/10}{1-1/10} = \frac{0.9}{0.9} = 1$$

1.1.2 Proof of General Case

Let the two representations be $0.a_1a_2\ldots$ and $0.b_1b_2\ldots$. Let k be the first index where the digits differ, i.e., $a_i = b_i$ for $i < k$, and $a_k \neq b_k$. Without loss of generality, assume $a_k < b_k$.

Since they represent the same number, the "smaller" starting sequence must be maximized in its tail, and the "larger" starting sequence must be minimized in its tail to meet.

1. **Maximize a-sequence**: The largest possible value starting with a_k is $0.a_1\ldots a_k999\ldots$.

2. **Minimize b-sequence**: The smallest possible value starting with b_k is $0.b_1 \dots b_k 000 \dots$.

For equality to hold:

$$0.a_1 \dots a_k 99 \cdots = 0.a_1 \dots b_k 00 \dots$$

This implies $b_k = a_k + 1$, and the tails must be strictly infinite 9s and infinite 0s respectively.

1.2 Periodic Decimals and Rational Numbers

Definition: A periodic decimal is a number of the form:

$$0.a_1 \dots a_k(c_1 \dots c_m) \quad \text{or} \quad 0.a_1 \dots a_k\overline{c_1 \dots c_m}$$

where $c_1 \dots c_m$ is the repeating period.

Examples:

- $\frac{1}{3} = 0.333 \cdots = 0.\overline{3}$
- $\frac{10}{7} = 1.428571428571 \cdots = 1.\overline{428571}$

Counter-example (Irrational):

$$0.12345678910111213 \dots$$

$$0.101001000100001 \dots$$

These do not repeat periodically.

1.2.1 Theorem 1: Rational $\implies$ Periodic

Any rational number has a periodic decimal representation.

Proof: Let $x = \frac{m}{n} \in \mathbb{Q}$. Perform long division of m by n. At each step, we calculate a remainder r_i. The possible values for any remainder are $\{0, 1, \dots, n-1\}$. Since there are only n distinct possible remainders, by the Pigeonhole Principle, a remainder must eventually repeat. Once a remainder repeats, the sequence of digits produced in the quotient will also repeat cyclically.

1.2.2 Theorem 2: Periodic $\implies$ Rational

If a real number has a periodic decimal representation, then it is rational.

Proof: Let $\Gamma = 0.a_1 \dots a_k \overline{c_1 \dots c_m}$. We can split this into a finite (non-periodic) part and an infinite (periodic) part.

$$\Gamma = \underbrace{0.a_1 \dots a_k}_{\text{Part I}} + \underbrace{0.00 \dots 0\overline{c_1 \dots c_m}}_{\text{Part II}}$$

1. **Part I**: This is a terminating decimal, which is clearly rational (Q).

2. **Part II**: This can be written as a geometric series.

   $$\text{Part II} = 10^{-k} \cdot (0.\overline{c_1 \dots c_m})$$

   Let $P = c_1 \dots c_m$ be the integer value of the period block.

   $$0.\overline{c_1 \dots c_m} = \frac{P}{10^m} + \frac{P}{10^{2m}} + \dots = P \sum_{j=1}^{\infty} (10^{-m})^j$$

   This is a convergent geometric series with ratio $r = 10^{-m} < 1$, which sums to a rational value.

Since Γ is the sum of two rationals, $\Gamma \in \mathbb{Q}$.

2 Numeral Systems

2.1 Positional Systems

A number n can be represented in base b ($b > 1$) as:

$$n = (a_k a_{k-1} \dots a_1 a_0)_b = \sum_{i=0}^{k} a_i \cdot b^i$$

where coefficients $a_i \in \{0, 1, \dots, b-1\}$.

2.1.1 Examples

- **Binary** ($b = 2$):

  $$(10011)_2 = 1 \cdot 2^4 + 0 \cdot 2^3 + 0 \cdot 2^2 + 1 \cdot 2^1 + 1 \cdot 2^0 = 16 + 2 + 1 = 19$$

- **Hexadecimal** ($b = 16$): Digits $0 - 9, A - F$.

  $$(A21)_{16} = 10 \cdot 16^2 + 2 \cdot 16^1 + 1 \cdot 16^0$$

2.1.2 Theorem (Existence and Uniqueness)

Let $b \in \mathbb{N}, b > 1$. Then any $n \in \mathbb{N}$ can be uniquely represented in base b.

2.2 Non-Positional Systems

Example: Roman Numerals (e.g., MMXXV). The value depends on fixed symbols rather than position powers (though order matters for subtraction/addition).

3 Number Theory Basics

3.1 Divisibility and The Division Algorithm

Definition: Let $a, b \in \mathbb{Z}$. We say a divides b (notation $a \mid b$) if there exists $c \in \mathbb{Z}$ such that $b = a \cdot c$.

Theorem (Division Algorithm): Let $a, b \in \mathbb{Z}$ with $a > 0$. There exist unique integers q (quotient) and r (remainder) such that:

$$b = a \cdot q + r, \quad 0 \leq r < a$$

3.2 Prime Numbers

Definition: A number $p > 1$ is prime if it has exactly two divisors: 1 and p.

- 101? Prime (YES).
- 57? Not Prime ($57 = 3 \times 19$, NO).

3.3 Greatest Common Divisor (GCD)

Definition: Let $a, b \in \mathbb{Z}$. A number c is a common divisor if $c \mid a$ and $c \mid b$. The **Greatest Common Divisor** (GCD), denoted $\gcd(a, b)$, is the largest integer that divides both a and b.

Examples:

- Factors of 6: 1, 2, 3, 6.
- Factors of 10: 1, 2, 5, 10.
- $\gcd(6, 10) = 2$.
- $\gcd(8, 16) = 8$.
- $\gcd(2, 5) = 1$ (Relatively prime).

4 Euclid's Algorithm

A method to compute $\gcd(a, b)$.

4.1 The Procedure

To find $\gcd(a, b)$:

1. $a = q_1 b + r_1$ where $0 \leq r_1 < b$
2. $b = q_2 r_1 + r_2$ where $0 \leq r_2 < r_1$
3. $r_1 = q_3 r_2 + r_3$
4. $\ldots$
5. $r_{n-2} = q_n r_{n-1} + r_n$
6. $r_{n-1} = q_{n+1} r_n + 0$

The algorithm terminates when the remainder is 0. The GCD is the **last non-zero remainder** r_n.

4.1.1 Example

Find $\gcd(49, 21)$.

1. $49 = 2 \cdot 21 + 7$
2. $21 = 3 \cdot 7 + 0$

$\implies \gcd(49, 21) = 7$.

4.2 Correctness Lemma

Lemma: Let $a = bq + r$. Then $\gcd(a, b) = \gcd(b, r)$.

Proof: Let S_1 be the set of common divisors of a and b. Let S_2 be the set of common divisors of b and r.

1. If $d \in S_1$, then $d \mid a$ and $d \mid b$. Since $r = a - bq$, d must divide linear combination $a - bq$, so $d \mid r$. Thus $d \in S_2$.
2. If $d \in S_2$, then $d \mid b$ and $d \mid r$. Since $a = bq + r$, d must divide $bq + r$, so $d \mid a$. Thus $d \in S_1$.

Since the sets of common divisors are identical, their greatest elements must be identical.

5 Bézout's Identity (Extended Euclidean Algorithm)

5.1 Theorem

Let $a, b \in \mathbb{N}$. Then there exist integers $x, y \in \mathbb{Z}$ such that:

$$\gcd(a, b) = a \cdot x + b \cdot y$$

5.1.1 Example

For $\gcd(49, 21) = 7$:

$$7 = 49(1) + 21(-2)$$

Here $x = 1, y = -2$.

5.2 Proof 1: Constructive (Back-substitution)

We can express the remainders in Euclid's algorithm linearly. From the algorithm:

$$r_i = r_{i-2} - q_i r_{i-1}$$

If previous remainders are linear combinations of a and b, then r_i is also a linear combination. By induction, since $r_1 = a - q_1 b$ (linear combination), all subsequent remainders, including $\gcd(a, b)$, are linear combinations of a and b.

5.3 Proof 2: Non-constructive (Well-Ordering Principle)

Consider the set of all positive linear combinations:

$$S = \{ax + by \mid x, y \in \mathbb{Z}, ax + by > 0\}$$

Let $m = \min(S)$. By the Well-Ordering Principle, a minimum exists. We claim $m = \gcd(a, b)$.

1. **Proof that m is a common divisor**: Suppose m does not divide a. By the Division Algorithm, $a = qm + r$ with $0 < r < m$.

   $$r = a - qm = a - q(ax + by) = a(1 - qx) + b(-qy)$$

   This implies r is a linear combination of a and b. Since $r > 0$, $r \in S$. But $r < m$, which contradicts that m is the minimum of S. Thus $m \mid a$. Similarly, $m \mid b$.

2. **Proof that m is the greatest**: Let d be any common divisor of a and b. Since $m = ax + by$ and d divides both a and b, d must divide $ax + by$, so $d \mid m$. Therefore, $d \leq m$.

Thus $m = \gcd(a, b)$.